A Visual-World Study of Processing Hungarian Pre-Verbal Focus Constructions

How languages express contrast and exhaustivity has major implications for how we understand the syntax-semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eyetracking with the visual-world paradigm. Specifically, we examine two theoretically important inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain *exhaustively*, and whether *contrastive but non-exhaustive* readings exist, as well as from a cognitively perspective, whether different focus types produce processing differences during language comprehension.

Constituents in canonical sentences can be focused for various communicative purposes, such as emphasis, question answering, and contradiction. For example, if one wants to focus the object of (1a) (see page 2), sentence (1b) should be used as a typical focus, called **pre-verbal focus (PVF)**, in which the focused element is moved to a pre-verbal position, and it often bears stress (Kornai & Kálmán, 1988). An intuitive reading of (1b) could be "*Anna ate the BANANA [and nothing else].*"; in this sense, PVF sentences seem to be equal to (1c), where the exhaustive interpretation is explicitly expressed with an operator *csak* 'only'. However, we also notice that if the utterance (1b) is amended with certain phrases, e.g. *többek között* 'among others' (1d), it is not only perfectly grammatical, but also expresses a contrastive sense. In Hungarian linguistic work, <u>semantic exhaustivity as a feature of PVF has been debated in the literature for over 40 years</u>. A fundamental question is: *does the exhaustivity arises from the syntax-semantic interface, or from pragmatic/contextual factors*? Within the generative framework, several studies argue that exhaustivity is an inherent feature of the PVF structure (Szabolcsi, 1981; É. Kiss, 1998; Kenesei, 2006), others advocate that exhaustive interpretation emerged from discourse-pragmatics, or conversational implicature (Wedgwood et al., 2006; Balogh, 2013, Káldi et al., 2016).

To address this theoretical debate, and in light with recent experimental studies (Onea & Beaver, 2009; Gerőcs et al., 2014; Káldi, 2021), we designed a visual-world eye-tracking experiment (n = 30; MeanAge = 33.5, female = 18) to differentiate interpretations associated with the PVF structures. Specifically, we were curious about Hungarian speakers' dominant interpretation of the unmodified, simple PVF sentence (e.g., (1b)), in comparison with the *only*-exhaustive (1c) and the contrastive condition (1d). We used a Sentence–Picture Verification (SPV) task (Fig. 1), in which participants read a sentence in a self-paced reading (SPR) manner and were asked to indicate one correct meaning-representation between two visual stimuli. Materials constituted 54 sentences (18 items × 3 conditions), and 54 pairs of images with relevant single and multiple objects (Fig. 2), and 54 filler trials. Using PCIbex (Schwarz & Zehr, 2021), we collected reading times (RT) during SPR, and eye-tracking data (total dwell time) during the SPV choice, recording the time spent looking at each target image.

The behavioural results showed that participants overwhelmingly interpreted the unmodified PVF sentences like *only*-sentences, choosing type A images, but in the "contrastive" condition (*among other*-sentences) they preferred the multi-object type B images, exhibiting a tendency toward non-exhaustive readings (Table 1). To further examine how the focus structures were associated with exhaustive and contrastive readings, we fitted a linear mixed-effects model to the total dwell time on images, with FocusCondition, ChoiceType and their interaction as main effects, and participant and item as random effects. The results revealed significant effects of FocusCondition ($\chi^2(2) = 11.67$, $p < .001$), ChoiceType ($\chi^2(1) = 19.51$, $p < .001$), and interaction between FocusCondition and ChoiceType ($p < .001$). Interestingly, post-hoc (TukeyHSD) tests showed that after reading *only*-sentences, participants took significantly less time to choose A compared to B ($\beta = -0.88$, $p < .001$), whereas they spent significantly more time to select A after reading contrastive sentences ($\beta = .21$, $p = .034$), suggesting that participants were faster to pick the expected choices in the "exhaustive" and "contrastive" conditions, respectively. Crucially, no significant difference was observed between selecting A or B for <u>the simple PVF sentences (β = -.19, p = .362), implying that participants might consider both representations equally plausible when making their choices</u>.

Overall, our results suggest that Hungarian pre-verbal focus constructions tend to be interpreted exhaustively when no additional context is provided. This exhaustive bias seems to persist even when 'among others' is present, despite its licensing a contrastive reading in truth value. This explains the apparent tension in theoretical accounts. Our results also showed facilitation for explicitly exhaustive sentences and not for simple PVFs, consistent with online disambiguation of an underspecified structure.

(1)	a	Anna		megette	a banánt.
		Anna.NOM		VPX-eat.PST3SG	the banana.ACC
		‘Anna ate the banana.’			
	b	Anna		**a BANÁNT**	ette meg.
		Anna.NOM		the banana.ACC	eat.PST3SG VPX
		‘Anna ate the BANANA.’			
	c	Anna	**csak**	**a BANÁNT**	ette meg.
		Anna.NOM	only	the banana.ACC	eat.PST3SG VPX
		‘Anna ate only the BANANA.’			
	d	Anna	**többek között**	**a BANÁNT**	ette meg.
		Anna.NOM	among others	the banana.ACC	eat.PST3SG VPX
		‘Anna ate the BANANA, among other things.’			

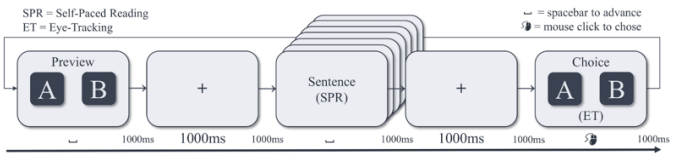

Figure 1. The design and procedure of the Sentence–Picture Verification task.

A B

Figure 2. Single-(A) and multi-object (B) target.

Condition	A % (N)	B % (N)
Exhaustive	93.9% (169)	6.1% (11)
PVF	94.4% (170)	5.6% (10)
Contrastive	41.7% (75)	58.3% (105)

Table 1. Type of chosen image per condition.

Condition	A Mean (SD)	B Mean (SD)
Exhaustive	1054 (1154)	2173 (2001)
PVF	1015 (991)	1844 (1939)
Contrastive	1206 (1323)	1226 (1157)

Table 2. Total dwell time during choice (ms).

References

Balogh, K. (2013). Hungarian Pre-verbal Focus and Exhaustivity. In Y. Motomura, A. Butler, & D. Bekki (Eds.), *New Frontiers in Artificial Intelligence: JSAI-isAI 2012 Workshops, LENLS, JURISIN, MiMI, Miyazaki, Japan, November 30 and December 1, 2012, Revised Selected Papers* (Vol. 7856, pp. 1–16). Springer. https://doi.org/10.1007/978-3-642-39931-2

É. Kiss, K. (1998). Identificational Focus versus Information Focus. *Language*, *74*(2), 245–273.

Gerőcs, M., Babarczy, A., & Surányi, B. (2014). *Exhaustivity in focus: Experimental evidence from Hungarian*. 181–195.

Káldi, T., Babarczy, A., & Bende-Farkas, Á. (2016). Hungarian focus: Presuppositional content and exhaustivity revisited. *Joseph E. Emonds–Markéta Janebová (szerk.) Language Use and Linguistic Structure. Proceedings of the Olomouc Linguistics Colloquium*, 245–262.

Káldi, T. (2021). *Hungarian Pre-verbal Focus: Representation and Interpretation* [Doctoral dissertation, Budapest University of Technology and Economics].

Kenesei, I. (2006). Focus as identification. In V. Molnár & S. Winkler (Eds.), *The Architecture of Focus,* 137-168. Walter de Gruyter.

Kornai, A., & Kálmán, L. (1988). Hungarian sentence intonation. In H. van der Hulst & N. Smith (Eds.), *Autosegmental studies on pitch accent* (pp. 183–195). Foris.

Mády, K. (2015). Prosodic (non-) realisation of broad, narrow and contrastive focus in Hungarian: A production and a perception study. *Speech Beyond Speech Towards a Better Understanding of the Most Important Biosignal: Proceedings of the Annual Conference of the International Speech Communication Association*, 948–952.

Onea, E., & Beaver, D. (2009). Hungarian focus is not exhausted. *Proceedings of Semantics and Linguistic Theory (SALT 19)*, 342–359. https://doi.org/10.3765/salt.v19i0.2524

Wedgwood, D., Petho, G., & Cann, R. (2006). *Hungarian ‘focus position’ and English it-clefts: The semantic underspecification of ‘focus’ readings*. [Manuscript, University of Edinburgh].